

ASSIGNMENT No: 8

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TITLE: Write a Java program to demonstrate the use of the super keyword in inheritance.

Problem Statement: Write a Java program to demonstrate the use of the super keyword in inheritance.

Learning Objectives: To demonstrate the use of the super keyword for accessing parent class members.

Theory:

The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class constructors, access parent class variables, and call overridden methods. Super helps distinguish between parent and child class members with the same name. It ensures proper constructor chaining during object creation. It is an important concept in inheritance and method overriding.

Exercise

1. Create an Employee and Manager program where Manager accesses Employee details using super keyword.

```
J Employee.java > Manager > main(String[])
1  class Employee {
2      String name = "Rahul";
3      int empId = 101;
4  }
5
6  class Manager extends Employee {
7
8      void display() {
9          System.out.println("Employee Name: " + super.name);
10         System.out.println("Employee ID: " + super.empId);
11     }
12
13     Run | Debug
14     public static void main(String[] args) {
15         Manager m = new Manager();
16         m.display();
17     }
18 }
```

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\arhaa\OneDrive\Desktop\java> & 'C:\Program Files\Java\
ng\Code\User\workspaceStorage\02e7f2cfce92a0ba8e30606ade6b2c24\redh
Employee Name: Rahul
Employee ID: 101
PS C:\Users\arhaa\OneDrive\Desktop\java>
```

2. Develop a Vehicle Insurance System where child insurance classes access parent vehicle information using the super keyword.

```
J Vehicle.java > Vehicle > main(String[])
C:\Users\arhaa\OneDrive\Desktop\java\Vehicle.java
2
3     String name = "Car";
4     String model = "Toyota";
5
6     // Inner Class
7     class VehicleDetails {
8         void display() {
9             System.out.println("Vehicle Name: " + name);
10            System.out.println("Vehicle Model: " + model);
11        }
12    }
13
14    Run | Debug
15    public static void main(String[] args) {
16
17        Vehicle v = new Vehicle();
18
19        // Inner Class Object
20        VehicleDetails vd = v.new VehicleDetails();
21        vd.display();
22
23        // Anonymous Class
24        Runnable action = new Runnable() {
25            public void run() {
26                System.out.println(x: "Vehicle is Running...");
27            }
28        };
29        action.run();
30    }
31
```

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
ng\Code\User\workspaceStorage\02e7f2cfce92a0ba8e30606ade6b2c24\redhat.java
Vehicle Name: Car
Vehicle Model: Toyota
Vehicle is Running...
PS C:\Users\arhaa\OneDrive\Desktop\java> ^C
PS C:\Users\arhaa\OneDrive\Desktop\java>
PS C:\Users\arhaa\OneDrive\Desktop\java> c:; cd 'c:\Users\arhaa\OneDrive\
onMessages' '-cp' 'C:\Users\arhaa\AppData\Roaming\Code\User\workspaceStora
Vehicle Name: Car
Vehicle Model: Toyota
Vehicle is Running...
PS C:\Users\arhaa\OneDrive\Desktop\java>
```